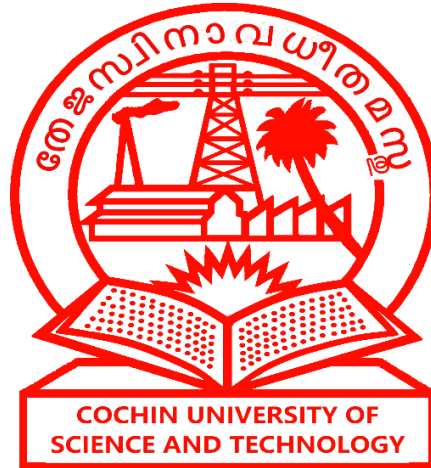


**COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
COCHIN UNIVERSITY COLLEGE OF ENGINEERING
KUTTANADU**



APRIL 2026

**PROJECT REPORT ON
SMART EV CHARGING SYSTEM**

Submitted on partial fulfillment of the requirement for the award of the degree in Master of
Computer Applications from COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY

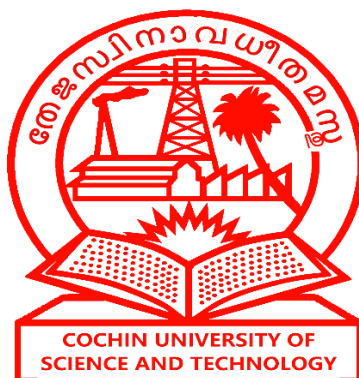
Submitted by

AMARNATH K (24100377)

DIVISION OF COMPUTER APPLICATIONS

2024-2026

COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
COCHIN UNIVERSITY COLLEGE OF ENGINEERING
KUTTANADU



CERTIFICATE

This is to certify that this project report entitled “**Smart EV Charging System**” is a bonafide record on partial fulfilment for the Degree of the Master of Computer Applications to the COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY through DIVISION OF COMPUTER APPLICATIONS, COCHIN UNIVERSITY COLLEGE OF ENGINEERING KUTTANADU, ALAPPUZHA done by **AMARNATH K (24100377)** in the year 2026.

Project Guide

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Asst Professor, MCA,
CUCEK

Project Coordinator

Ms DIVYA G
Asst Professor, MCA,
CUCEK

Head of the Department

Dr. PREETHA MATHEW K
Principal
CUCEK

DECLARATION

I hereby declare that the project entitled “**Smart EV Charging System**” submitted to the DIVISION OF COMPUTER APPLICATIONS, COCHIN UNIVERSITY COLLEGE OF ENGINEERING, KUTTANADU in the partial fulfilment of the requirements for the award of Degree in MASTER OF COMPUTER APPLICATIONS is a record of original work done by me under the guidance of **Ms. RIYA RASHEED** , Assistant Professor in MCA Department during my period of studying, Cochin University College of Engineering, Kuttanadu.

Place :

AMARNATH K

Date :

ACKNOWLEDGEMENT

I am thankful to God almighty for the blessings in the successful completion of my main project “**Smart EV Charging System**”. I would like to record my profound gratitude to **Dr. PREETHA MATHEW K**, Principal and Head of the Department, MCA, Cochin University College of Engineering who has deeply inspired me to do our project.

It's grateful to express my thanks to **Ms. RIYA RASHEED**, Assistant Professor in MCA Department my Project guide Cochin University College of Engineering, Kuttanadu, because of her effective guidance, constructive criticism and innovative and useful stream of suggestions that helped me to complete my project.

I am also grateful to my Project coordinator **Ms.DIVYA G**, Associate Professor of DCA department for providing various resources and moral support throughout the project development period .I am thankful to various resources that provide requirements for my project, because requirements are backbone of every project. I am also thankful to our teachers, friends, family members, for their support and prayer for me to complete my project.

ABSTRACT

The **Smart EV Charging System** is a web-based application designed to improve the efficiency and convenience of electric vehicle (EV) charging. With the rapid increase in the number of electric vehicles, managing charging stations has become a significant challenge due to issues such as long waiting times, poor slot management, and lack of real-time information.

This system provides an organized platform where users can register, log in, search for nearby charging stations, view available charging slots, and book slots in advance. Once a booking is made, a QR code is generated for verification at the charging station, ensuring secure and efficient access. Users can also view their booking history and cancel bookings when required.

The system is developed using Node.js and Express.js for the backend, MySQL as the database, and HTML, CSS, and JavaScript for the frontend. It follows a client-server architecture and implements secure authentication using JWT.

The main objective of this project is to reduce waiting time, optimize slot utilization, and enhance user convenience. By providing a real-time and automated booking system, the application improves the overall management of EV charging stations and contributes to a more efficient and sustainable transportation ecosystem.

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INTRODUCTION

1.1 ABOUT THE PROJECT

The **Smart EV Charging System** is a web-based platform designed to manage electric vehicle charging stations and booking operations in an organized manner. With the rapid growth of electric vehicles, managing charging stations efficiently has become a major challenge. Many users face issues such as long waiting times, lack of slot availability information, and poor coordination between users and service providers.

This project provides a centralized system where users can search for charging stations, view available slots, and book charging sessions in advance. The system generates a QR code for each booking, which can be used for verification at the charging station. It also allows users to view booking history and cancel bookings when necessary.

The system includes features such as user registration, login, station viewing, slot management, booking, QR code generation, and booking tracking. The interface is designed to be simple and user-friendly so that users can easily interact with the system.

Developed using modern web technologies, the system ensures efficient data management, easy accessibility, and improved coordination between users and charging station providers. Overall, the project enhances convenience, reduces waiting time, and improves the overall EV charging experience.

1.2 OBJECTIVE & SCOPE OF THE PROJECT

OBJECTIVE

The main objective of the Smart EV Charging System is to provide a structured and efficient way to manage EV charging operations digitally. The system aims to reduce waiting time and improve slot management by allowing users to book charging slots in advance.

It focuses on maintaining accurate data related to users, charging stations, available slots, and bookings in a centralized database. The system also ensures secure access through authentication and enables users to manage their bookings efficiently.

By providing a simple interface and real-time data handling, the system improves user convenience and supports better utilization of charging infrastructure.

SCOPE

The main objective of the Smart EV Charging System is to provide a structured and efficient way to manage EV charging operations digitally. The system aims to reduce waiting time and improve slot management by allowing users to book charging slots in advance.

It focuses on maintaining accurate data related to users, charging stations, available slots, and bookings in a centralized database. The system also ensures secure access through authentication and enables users to manage their bookings efficiently.

By providing a simple interface and real-time data handling, the system improves user convenience and supports better utilization of charging infrastructure.

1.3 DEFINITION OF PROBLEM:

In the current scenario, EV users often face difficulties in finding available charging stations and managing charging time efficiently. Many charging stations do not provide real-time information about slot availability, leading to overcrowding and long waiting times.

Users are required to physically visit stations to check availability, which results in inconvenience and inefficient use of resources. There is also a lack of proper communication between users and service providers.

The Smart EV Charging System addresses these issues by providing a digital platform where users can check availability, book slots in advance, and manage their charging sessions efficiently.

LITERATURE REVIEW

2. LITERATURE REVIEW

1. P. Palka, T. Śliwiński et al. (2025)

This study presents an advanced scheduling system for electric vehicle charging using optimization techniques and machine learning models. The system selects optimal charging stations along a route by considering factors such as travel time, battery status, and user preferences. It focuses on reducing waiting time and improving charging efficiency through intelligent scheduling. The results show improved execution performance and better decision-making for EV users. The study highlights the importance of optimized scheduling in EV charging systems.

2. S. Jaman et al. (2023)

This research proposes a smart EV charging station system integrated with renewable energy sources such as solar power and battery storage. The system includes an intelligent power management and scheduling mechanism to optimize charging demand and reduce energy costs. It dynamically allocates charging power based on user demand and available resources. The study demonstrates that smart scheduling improves energy efficiency and supports scalable EV charging infrastructure.

3. H. I. Shaheen et al. (2024)

This study focuses on optimizing EV charging and discharging using metaheuristic algorithms such as Particle Swarm Optimization and Whale Optimization Algorithm. The system supports Vehicle-to-Grid (V2G) technology, allowing EVs to interact with the power grid. The results show that optimized scheduling reduces charging costs and improves energy management. The study highlights the importance of intelligent algorithms in managing EV charging systems efficiently.

4. I. Chandra et al. (2025)

This research introduces an optimization-based approach for managing solar-powered EV charging stations. It uses advanced swarm optimization techniques to minimize charging costs and improve grid stability. The system evaluates different charging modes and demonstrates reduced power losses and improved efficiency. The study emphasizes the role of renewable energy integration in EV charging infrastructure.

5. S. Shao et al. (2022)

This study presents a scheduling model for managing multiple EV charging requests using dynamic programming techniques. It considers varying arrival times and charging requirements of vehicles. The system improves resource allocation and reduces congestion at charging stations. The research highlights the importance of intelligent scheduling algorithms in large-scale EV charging systems.

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

The existing system for EV charging is mostly manual or lacks proper digital integration. Users need to visit charging stations physically to check availability, and there is no centralized platform to manage bookings.

This results in inefficient use of charging stations and increased waiting time for users.

3.1.1 DISADVANTAGES

- No advance booking system
- Long waiting times at charging stations
- Lack of real-time slot availability
- Poor management of charging resources
- No centralized system for users

3.2 PROPOSED SYSTEM

The proposed system provides a web-based platform for managing EV charging stations and bookings efficiently. Users can register, log in, view stations, check available slots, and book slots in advance.

The system generates QR codes for booking verification and maintains all booking data in a centralized database.

3.2.1 ADVANTAGES

Reduces waiting time

- Provides real-time slot availability
- Enables advance booking

- Improves user convenience
- Secure booking with QR code
- Centralized data management

3.2.2 MODULES

- Registration
- Login
- View Charging Stations
- View Available Slots
- Book Slot
- View My Bookings
- Cancel Booking
- QR Code Generation
- Admin Management

3.3 FEASIBILITY STUDY

A feasibility study evaluates whether the proposed **Smart EV Charging System** can be successfully developed, executed, and sustained using the available resources. It examines the technical, economic, operational, and time-related factors to ensure that the project is practical, beneficial, and aligned with user needs. By analyzing each aspect, the study confirms that the system can function efficiently and deliver long-term improvements to hostel management processes.

1. Technical Feasibility: -

The project is technically feasible because it uses robust and widely supported technologies such as Node.js, Express.js, and MySQL, ensuring secure and scalable development.

- These technologies offer good community support, documentation, and regular updates.
- The system architecture is simple enough to implement yet efficient for handling large data volumes.

2. Economic Feasibility: -

The system is economically feasible as it minimizes development and maintenance costs by relying on open-source tools.

- No licensing cost
- Low maintenance cost
- Reduces operational expenses

3. Operational Feasibility: -

The system is operationally feasible because it provides a user-friendly interface that can be easily adopted by managers, owners, and administrators.

- Simple interface
- Easy navigation
- Improves efficiency

4. Time Feasibility: -

The project is time-feasible, as the system can be developed through a modular approach, allowing components to be built and tested individually.

- Faster development using frameworks
- Parallel module development possible

SYSTEM REQUIREMENTS
AND
SPECIFICATIONS

4.1 HARDWARE CONFIGURATIONS

The selection of hardware is very important in the existence and proper working of any of the software. When selecting hardware, the size and capacity requirements are also important. This software is able to run with following hardware configuration.

Processor	: Intel core i3
Processor Speed	: 1.5GHz to 2.2GHz
RAM	: 4GB or above
Hard Disk	: 3GB and above
Keyboard	: 108 keys
Clock speed	: 500 MHz
System bus	: 32 bits

4.2 SOFTWARE CONFIGURATIONS

One of the most difficult tasks is selecting software, once the system requirement is found out, then we have to determine whether a particular software package fits for those system requirements. This section summarizes the application requirement.

Operating System: Windows 7/8/8.1/10/11.

Language: Node.js with Express.js

IDE: VS Code

Database: SQLite

4.3 TECHNOLOGY USED

- Node.js
- Express.js
- MySQL Database
- HTML, CSS, JavaScript
- JWT Authentication
- QR Code Library

3.4 PLATFORM USED

- VS Code IDE
- Web Browser (Chrome/Edge)
- Windows / Linux

SYSTEM DESIGN

5.1 DATABASE DESIGN

The general theme behind a database is to handle information as an integrated whole. A database is a collection of interrelated data stored with minimum redundancy to serve many users quickly and effectively. After designing input and output, the analyst must concentrate on database design or how data should be organized around user requirements. The general objective is to make information access easy, quick, inexpensive and flexible for other users.

Database Schema

USERS

Column name	Datatype	Constraints	Description
Id	integer	primary key, autoincrement	unique identifier for each user
username	text	unique, not null	login username
Email	text	unique, not null	user email address
password	text	not null	stored encrypted password
created_date	text	default current_timestamp	date the account was created
Role	text	default 'user'	defines user role such as admin, staff, or customer

STATIONS

Column Name	Datatype	Constraints	Description
id	int	primary key, auto increment	unique station id
name	varchar	not null	station name
location	varchar	not null	station location
total_slots	int	default 0	total slots available
created_at	timestamp	default current_timestamp	station creation date

SLOTS

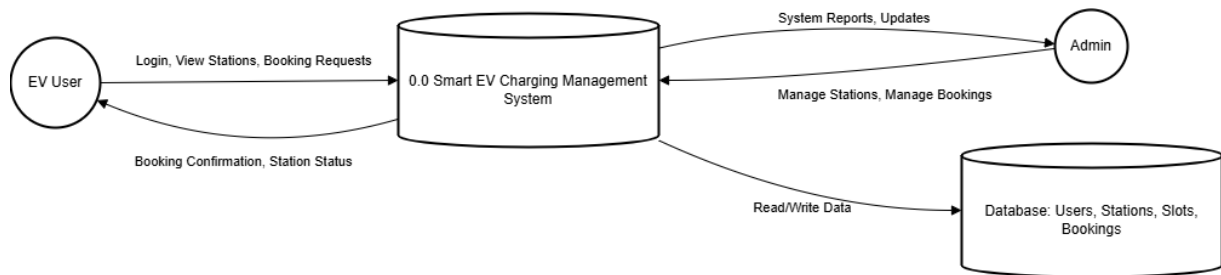
Column Name	Datatype	Constraints	Description
id	int	primary key, auto increment	unique station id
name	varchar	not null	station name
location	varchar	not null	station location
total_slots	int	default 0	total slots available
created_at	timestamp	default current_timestamp	station creation date

BOOKINGS

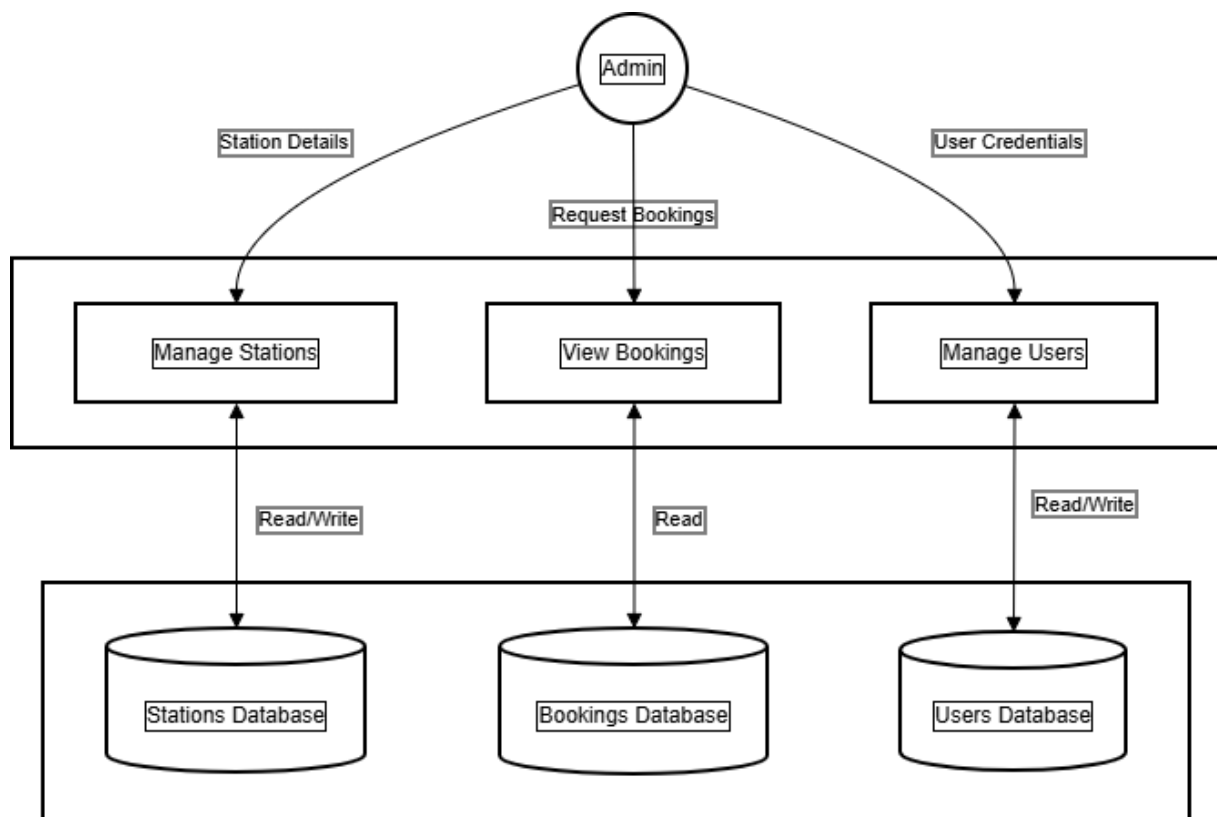
Column Name	Datatype	Constraints	Description
id	integer	primary key, autoincrement	unique purchase order id
item_id	integer	foreign key	linked item id
quantity	integer	not null	quantity ordered
status	text	default 'pending'	purchase progress status
created_at	timestamp	default current_timestamp	record creation date
received_at	timestamp	-	date the order was received

5.2 DATAFLOW DIAGRAM

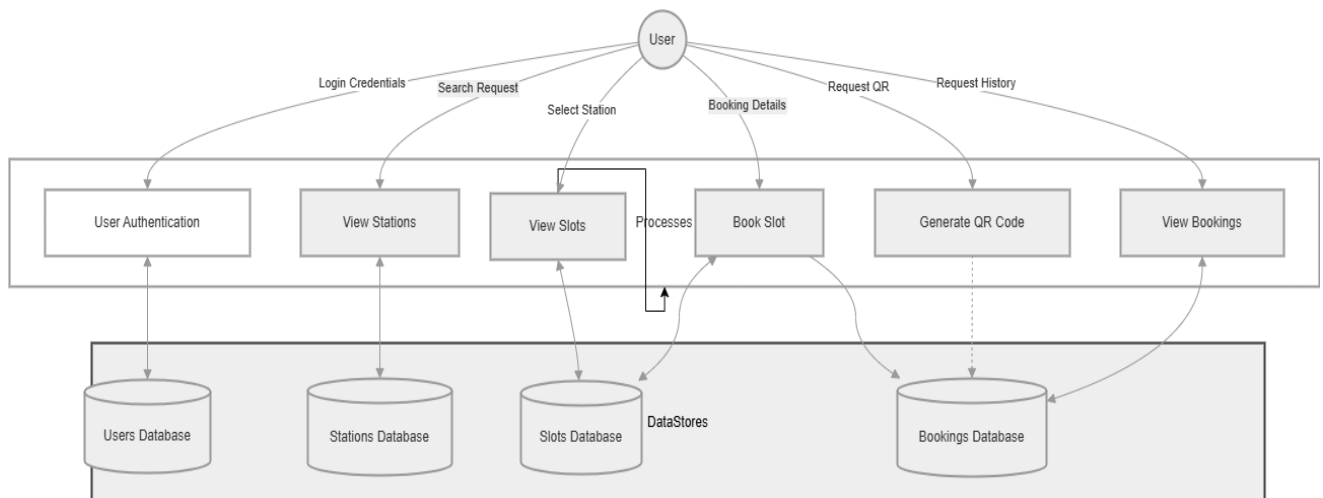
5.2.1 DFD LEVEL 0



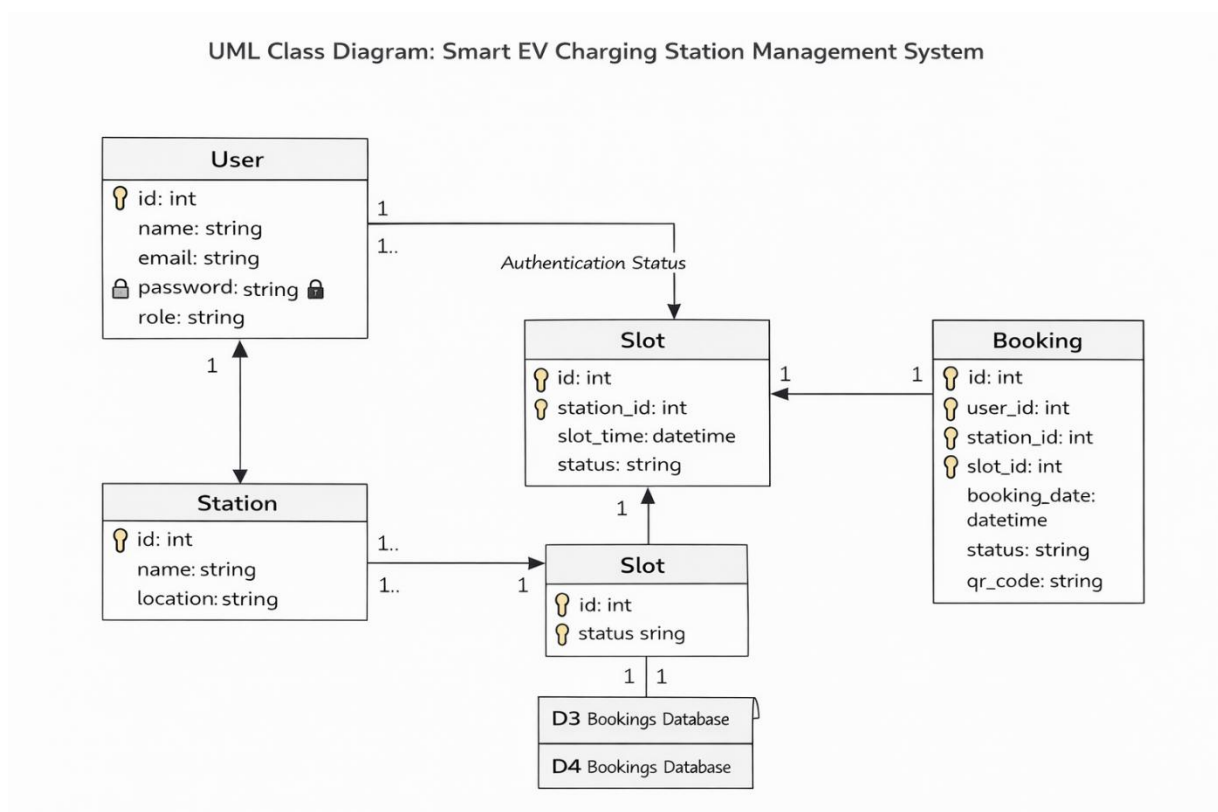
5.2.2 DFD LEVEL 1 ADMIN



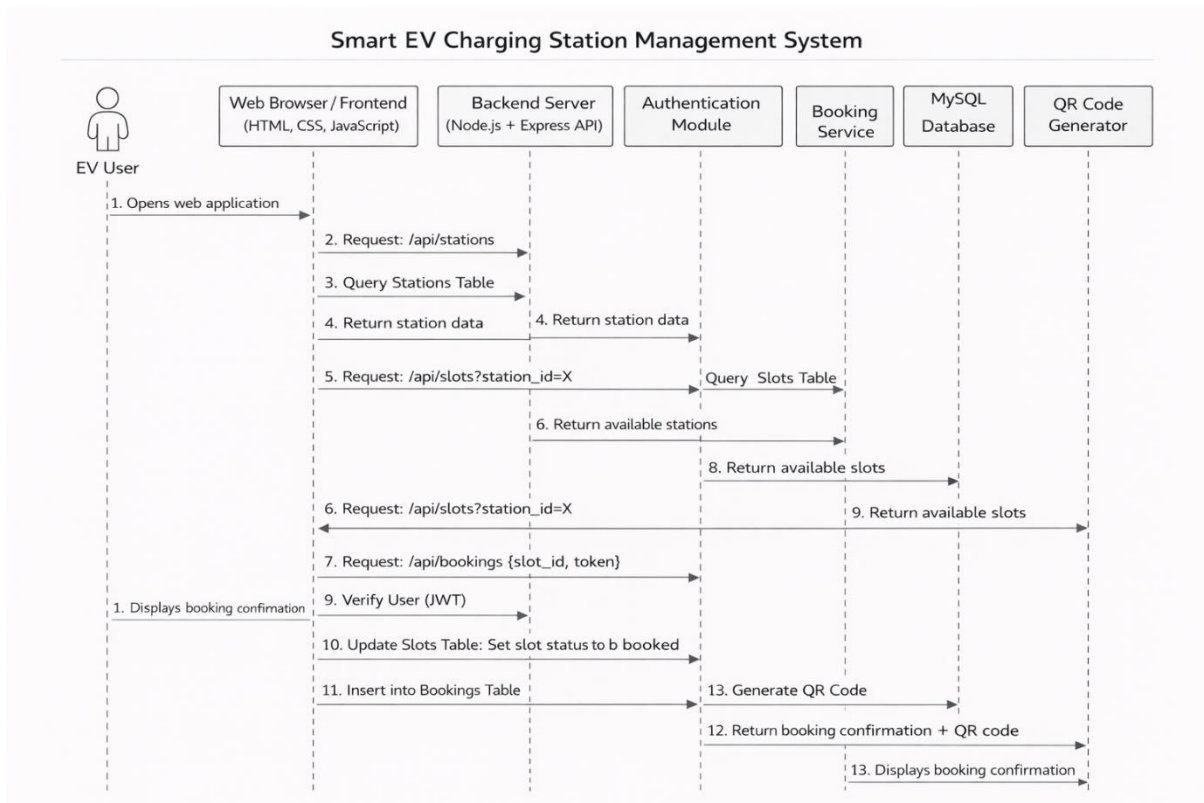
5.2.3 DFD LEVEL 1 CUSTOMER



5.3 CLASS DIAGRAM

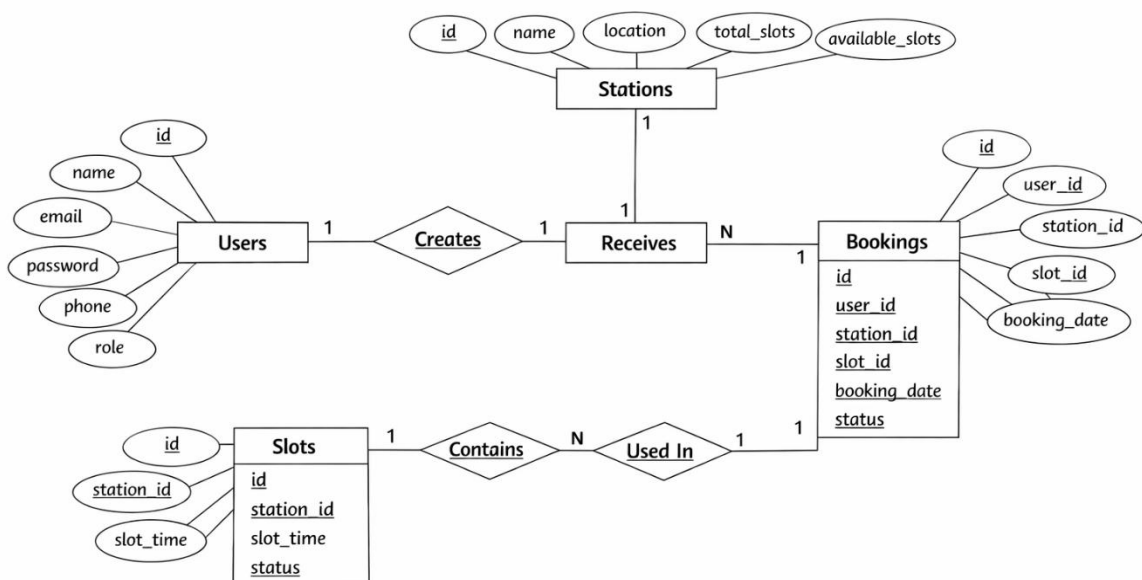


5.4 SEQUENCE DIAGRAM

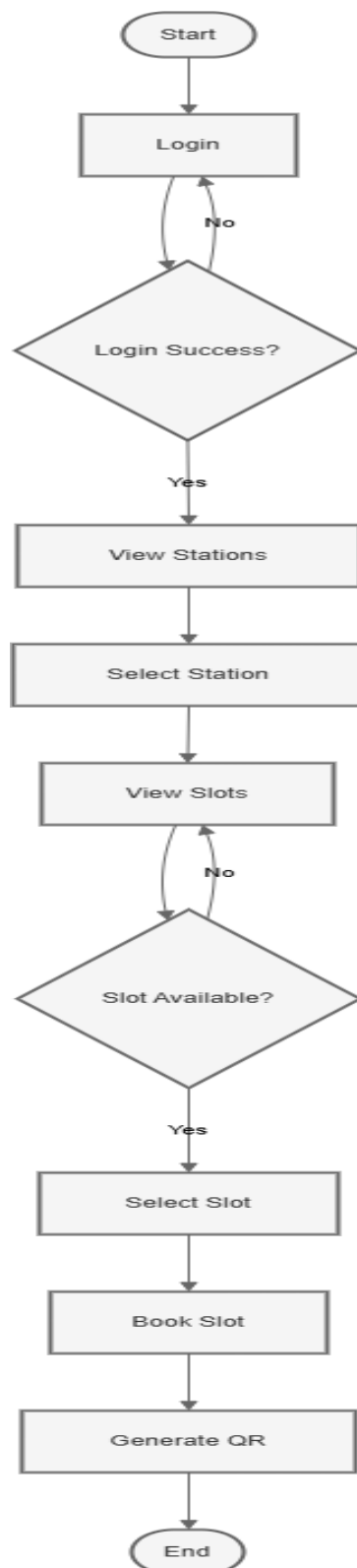


5.5 ER DIAGRAM

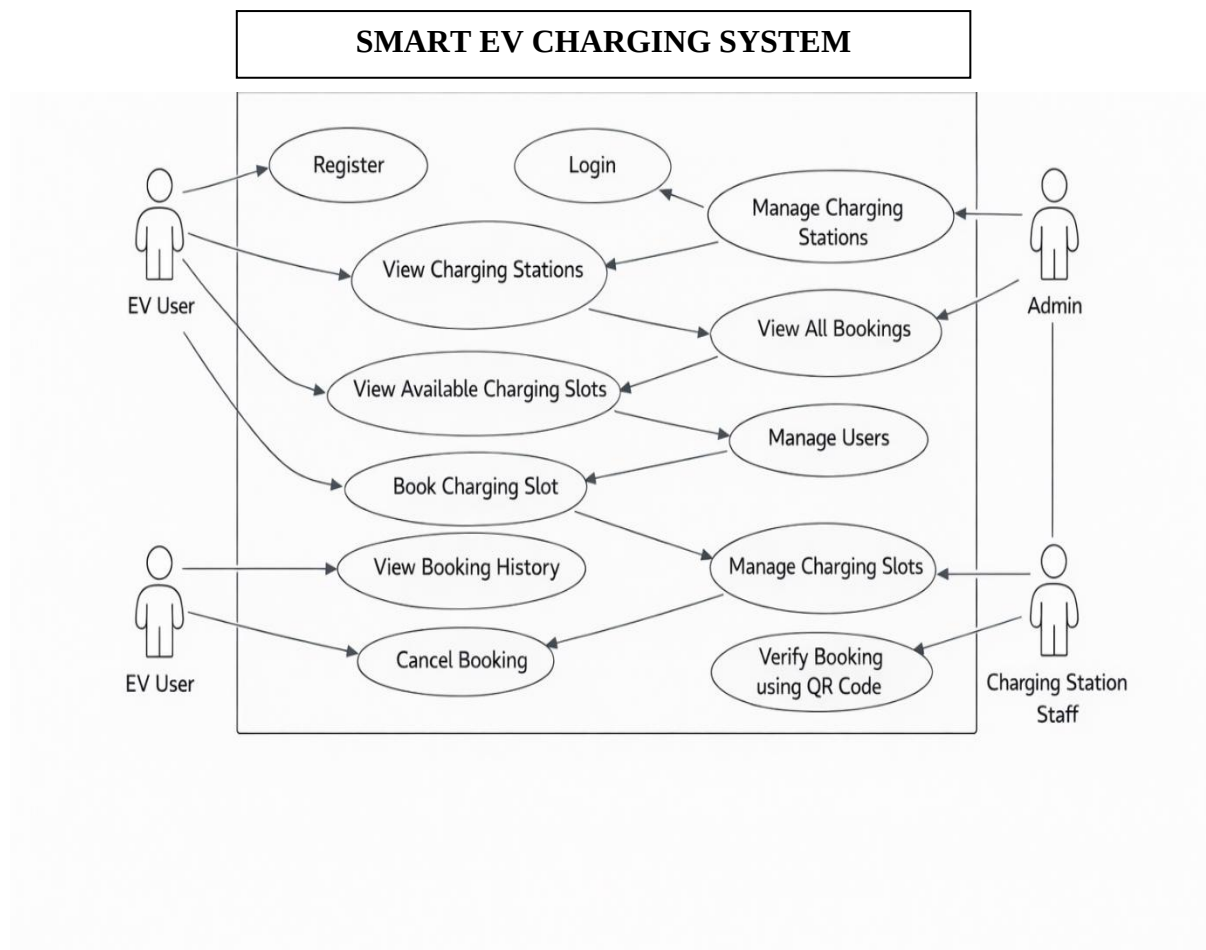
Entity Relationship Diagram – Smart EV Charging Station Management System



5.6 ACTIVITY DIAGRAM



5.7 USE CASE DIAGRAM



**SYSTEM IMPLEMENTATION,
TESTING
&
MAINTENANCE**

6.1 SYSTEM IMPLEMENTATION

We are implementing the system in widely supported technologies such as Node.js, Express.js, and MySQL, ensuring secure and scalable development.

6.2 SYSTEM TESTING

System testing is a level of testing that validates the complete and fully integrated software product. The purpose of a system test is to evaluate the end-to-end system specifications. Usually, the software is only one element of a larger computer-based system. Ultimately, the software interfaces with other software/hardware systems. System Testing is actually a series of different tests whose sole purpose is to exercise the full computer-based system.

Two Category of Software Testing

- Black Box Testing
- White Box Testing

System test falls under the **black box testing** category of software testing.

White box testing is the testing of the internal workings or code of a software application. In contrast, black box or System Testing is the opposite. System test involves the external workings of the software from the user's perspective.

System Testing involves testing the software code for following

- Testing the fully integrated applications including external peripherals in order to check how components interact with one another and with the system as a whole. This is also called an End-to-End testing scenario.
- Verify thorough testing of every input in the application to check for desired outputs.
- Testing of the user's experience with the application.

That is a very basic description of what is involved in system testing. You need to build detailed test cases and test suites that test each aspect of the application as seen from the outside without looking at the actual source code.

6.2.1 UNIT TESTING

Unit testing performed on each module or block of code during development. The programmer who writes the code normally does unit testing. It focuses on the smallest unit of software design. In this we test an individual unit or group of inter related units. Programmers often do it by using sample input and observing its corresponding outputs.

The main aim is to isolate each unit of the system to identify, analyse and fix the defects.

ADVANTAGES:

- Reduces Defects in the newly developed features or reduces bugs when changing the existing functionality.
- Reduces Cost of Testing as defects are captured in a very early phase.
- Improves design and allows better refactoring of code.
- Unit Tests, when integrated with build, give the quality of the build as well.

TECHNIQUES:

- **Black Box Testing** - Using which the user interface, input and output are tested.
- **White Box Testing** - used to test each one of those function's behavior is tested.
- **Grey Box Testing** - Used to execute tests, risks and assessment methods.

6.2.2 INTEGRATION TESTING

The meaning of Integration testing is quite straightforward- Integrate/combine the unit tested module one by one and test the behavior as a combined unit. The main function

or goal of this testing is to test the interfaces between the units/modules. We normally do Integration testing after “Unit testing”. Once all the individual units are created and tested, we start combining those “Unit Tested” modules and start doing the integrated testing. The main function or goal of this testing is to test the interfaces between the units/modules. The individual modules are first tested in isolation. Once the modules are unit tested, they are integrated one by one, until all the modules are integrated, to check the combinational behavior, and validate whether the requirements are implemented correctly or not. Here we should understand that Integration testing does not happen at the end of the cycle, rather it is conducted simultaneously with the development. So, most of the time all the modules are not actually available to test and here is what the challenge comes to test something which does not exist!

ADVANTAGES:

There are several advantages of this testing and few of them are listed below:

- This testing makes sure that the integrated modules/components work properly.
- Integration testing can be started once the modules to be tested are available. It does not require the other module to be completed for testing to be done, as Stubs and Drivers can be used for the same.
- It detects the errors related to the interface.

6.2.3 SYSTEM TESTING

System Testing (ST) is a black box testing technique performed to evaluate the complete system's compliance against specified requirements. In System testing, the functionalities of the system are tested from an end-to-end perspective. System Testing is usually carried out by a team that is independent of the development team in order to measure the quality of the system unbiased. It includes both functional and

NonFunctional testing. Black-box testing treats the software as a "black box", examining functionality without any knowledge of internal implementation, without seeing the source code. The testers are only aware of what the software is supposed to do, not how it does it. Black-box testing methods include: equivalence partitioning, boundary value analysis, all-pairs testing, state transition tables, decision table testing, fuzz testing, model-based testing, use case testing, exploratory testing and specification-based testing. Specification-based testing aims to test the functionality of software according to the applicable requirements. This level of testing usually requires thorough test cases to be provided to the tester, who then can simply verify that for a given input, the output value (or behavior), either "is" or "is not" the same as the expected value specified in the test case. Test cases are built around specifications and requirements, i.e., what the application is supposed to do. It uses external descriptions of the software, including specifications, requirements, and designs to derive test cases. These tests can be functional or non-functional, though usually functional. Specification-based testing may be necessary to assure correct functionality, but it is insufficient to guard against complex or high-risk situations. One advantage of the black box technique is that no programming knowledge is required. Whatever biases the programmers may have had, the tester likely has a different set and may emphasize different areas of functionality.

On the other hand, black-box testing has been said to be "like a walk in a dark labyrinth without a flashlight. "Because they do not examine the source code, there are situations when a tester writes many test cases to check something that could have been tested by only one test case, or leaves some parts of the program untested.

6.2.4 TEST REPORT

Test Report is needed to reflect testing results in a formal way, which gives an opportunity to estimate testing results quickly. It is a document that records data obtained from an evaluation experiment in an organized manner, describes the environmental or operating conditions, and shows the comparison of test results with test objectives.

6.3 SYSTEM MAINTENANCE

An integral part of software is the maintenance one, which requires an accurate maintenance plan to be prepared during the software development. It should specify how users will request modifications or report problems. The budget should include resource and cost estimates. A new decision should be addressed for the development of every new system feature and its quality objectives. The software maintenance, which can last for 5–6 years (or even decades) after the development process, calls for an effective plan which can address the scope of software maintenance, the tailoring of the post-delivery/deployment process, the designation of who will provide maintenance, and an estimate of the life-cycle costs. The selection of proper enforcement of standards is the challenging task right from the early stage of software engineering which has not got definite importance by the concerned stakeholders.

SAMPLE CODE

server.js

```
const express = require('express');

const cors = require('cors');

const app = express();

const db = require('./config/db');


// ★ JSON middleware MUST come BEFORE routes

app.use(cors());

app.use(express.json());


const authRoutes = require('./routes/authRoutes');

app.use('/api/auth', authRoutes);


const stationRoutes = require('./routes/stationRoutes');

app.use('/api/stations', stationRoutes);


const bookingRoutes = require('./routes/bookingRoutes');

app.use('/api/booking', bookingRoutes);


const adminRoutes = require('./routes/adminRoutes');

app.use('/api/admin', adminRoutes);


const staffRoutes = require('./routes/staffRoutes');

app.use('/api/staff', staffRoutes);
```



```
app.get('/', (req, res) => {  
  res.send("EV Charging Server Running 🚗⚡");  
});
```

```
app.listen(5000, () => {  
  console.log("Server started on port 5000");  
});
```

index.html

```
<!DOCTYPE html>  
  
<html lang="en">  
  
<head>  
  
  <meta charset="UTF-8">  
  
  <meta name="viewport" content="width=device-width, initial-scale=1.0">  
  
  <meta name="description" content="Smart EV Charging System - Find, book, and manage  
electric vehicle charging slots with ease.">  
  
  <title>Smart EV Charging</title>  
  
  <link rel="stylesheet" href="css/style.css">  
  
</head>  
  
<body>  
  
  <!-- Navbar -->  
  
  <div class="navbar">  
  
    <a href="/" class="logo">  
  
      <span class="logo-icon">⚡</span>
```

```
<span class="logo-text">EV<span>Charge</span></span>

</a>

<nav>

  <a href="login.html" class="nav-btn nav-btn-outline">Log In</a>

  <a href="register.html" class="nav-btn nav-btn-primary">Sign Up</a>

</nav>

</div>


<!-- Hero Section -->

<section class="hero">

  <div class="hero-content">

    <div class="hero-badge fade-in">

      <span class="badge-dot"></span>

      Live charging stations available

    </div>

    <h1 class="fade-in fade-in-delay-1">

      Smart <span class="highlight">EV Charging</span> Made Simple

    </h1>

    <p class="hero-desc fade-in fade-in-delay-2">

      Find nearby stations, book your slot in seconds, and get a QR code for seamless check-in.
      No more waiting — charge smarter.

    </p>

    <div class="hero-actions fade-in fade-in-delay-3">
```

```
<a href="register.html" class="btn btn-primary">Get Started →</a>

<a href="login.html" class="btn btn-outline">I have an account</a>

</div>

<div class="hero-stats fade-in fade-in-delay-3">

  <div class="hero-stat">

    <div class="stat-value">6+</div>

    <div class="stat-label">Stations</div>

  </div>

  <div class="hero-stat">

    <div class="stat-value">30+</div>

    <div class="stat-label">Slots</div>

  </div>

  <div class="hero-stat">

    <div class="stat-value">24/7</div>

    <div class="stat-label">Availability</div>

  </div>

</div>

</div>

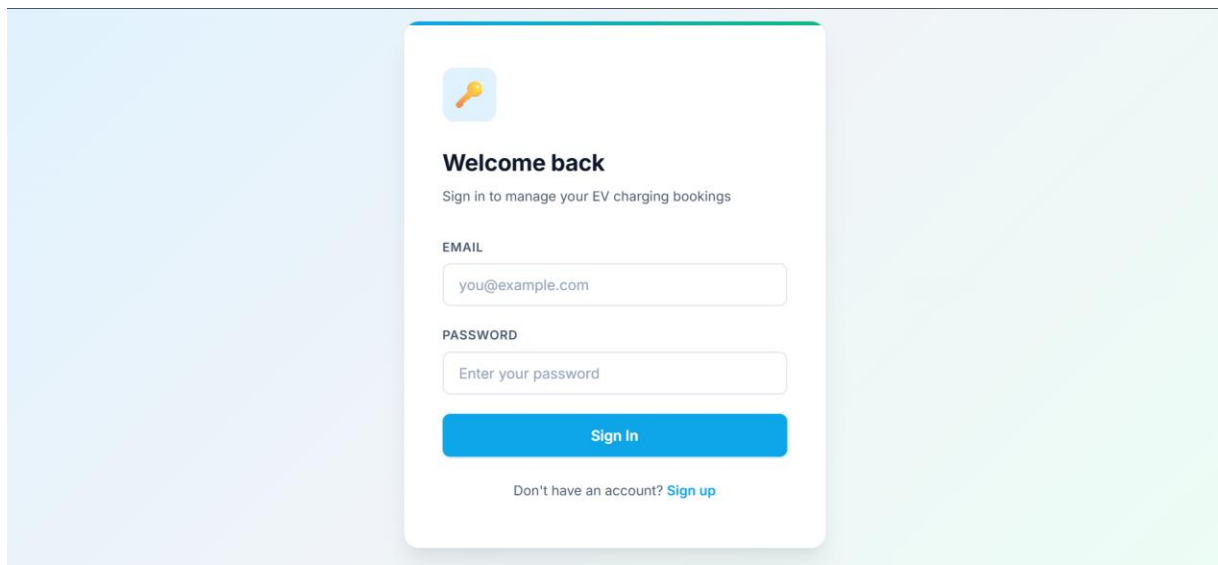
</section>

</body>

</html>
```

SCREENSHOTS

LOGIN



Welcome back

Sign in to manage your EV charging bookings

EMAIL

you@example.com

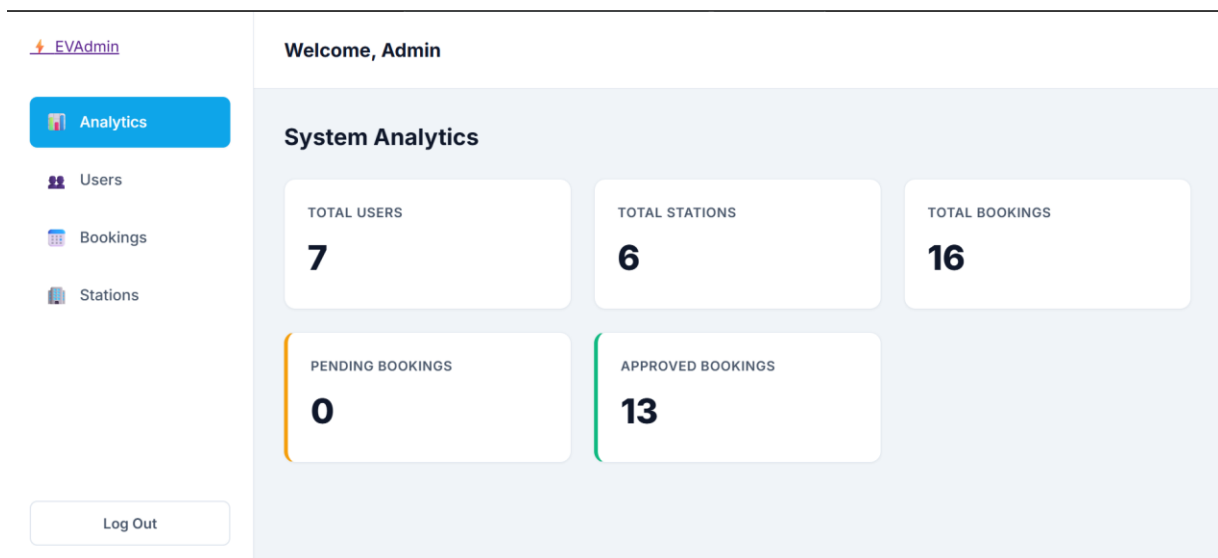
PASSWORD

Enter your password

Sign In

Don't have an account? [Sign up](#)

ADMIN PAGE



Welcome, Admin

System Analytics

Metric	Value
TOTAL USERS	7
TOTAL STATIONS	6
TOTAL BOOKINGS	16
PENDING BOOKINGS	0
APPROVED BOOKINGS	13

EVAdmin

- Analytics**
- Users
- Bookings
- Stations

Log Out

CUSTOMER PAGE



StationsMy BookingsLog Out

Charging Stations

Browse available stations and book a slot to charge your EV

Search



GreenCharge Hub

 MG Road, Bangalore

 Total: 5  Available: 5

View Available Slots →



PowerUp Station

 Anna Nagar, Chennai

 Total: 5  Available: 5

View Available Slots →



EcoVolt Center

 Banjara Hills, Hyderabad

 Total: 5  Available: 5

View Available Slots →

STAFF PAGE



Station DetailsManage SlotsManage Bookings

Log Out

Station Staff Dashboard

Your Assigned Station

Downtown Fast Charger

 123 Main St, City Center

Total Slots: 10Available Slots: 10

Edit Details

Activate Windows
Go to Settings to activate Windows.

 EVStaff Station Details Manage Slots Manage Bookings

Log Out

Station Staff Dashboard

Manage Charging Slots

[+ Add New Slot](#)

Add Charging Slot

SLOT TIME (E.G., 10:00 AM - 11:00 AM)

10:00 AM - 11:00 AM

STATUS

Available

[Add Slot](#)[Cancel](#)

ID	TIME	STATUS	ACTIONS
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Activate Windows

Go to Settings to activate Windows.

CONCLUSION

9.1 CONCLUSION

The **Smart EV Charging Station Management System** successfully provides a digital solution for managing electric vehicle charging operations efficiently. The system enables users to register, log in, view charging stations, check available slots, and book charging sessions in advance. The integration of QR code generation ensures secure and reliable verification at the charging station.

By replacing manual or unorganized methods with a structured web-based platform, the system reduces waiting time, improves slot management, and enhances overall user convenience. The use of modern technologies such as Node.js, Express.js, MySQL, and JavaScript ensures scalability, reliability, and efficient data handling.

The project meets its primary objectives by providing a centralized system that improves coordination between users and service providers. It also demonstrates the practical implementation of web technologies in solving real-world problems related to EV charging infrastructure.

9.2 FUTURE SCOPE

The current system provides a strong foundation for managing EV charging operations; however, several enhancements can be made to improve its functionality and usability in the future.

Future improvements may include:

- Integration of online payment systems for seamless booking transactions
- Development of a mobile application for better accessibility
- Real-time tracking of charging station availability
- Integration with GPS for location-based station search
- Implementation of automated slot allocation and smart scheduling
- Cloud-based deployment for better scalability and data access
- Advanced analytics and reporting for usage patterns and demand prediction
- Notification system for booking reminders and slot availability alerts

These enhancements will further improve system performance, user experience, and scalability, making it suitable for large-scale deployment in smart city environments.

REFERENCES

10. REFERENCES

- [1] P. Pałka, T. Śliwiński et al., “Planning and Scheduling System for Electric Vehicle Charging,” *ResearchGate*, 2025.
- [2] S. Jaman et al., “Smart Electric Vehicle Charging Station Integrated with Renewable Energy Sources,” *Sustainable Energy Technologies and Assessments*, 2023.
- [3] H. I. Shaheen et al., “Optimization Techniques for Electric Vehicle Charging and Discharging Systems,” *Energy Reports*, 2024.
- [4] J. Lee et al., “IoT-Based Smart Electric Vehicle Charging System,” *Energies*, vol. 14, no. 15, 2021.
- [5] D. Yang et al., “Real-Time Management of Electric Vehicle Fast Charging Stations,” *arXiv preprint arXiv:2012.09349*, 2020.
- [6] Node.js Foundation, “Node.js Documentation,” [Online]. Available: <https://nodejs.org>.
- [7] Express.js, “Express Web Framework Documentation,” [Online]. Available: <https://expressjs.com>.
- [8] Oracle Corporation, “MySQL Documentation,” [Online]. Available: <https://dev.mysql.com/doc>.
- [9] Mozilla Developer Network, “Web Technologies Documentation (HTML, CSS, JavaScript),” [Online]. Available: <https://developer.mozilla.org>.
- [10] Kazuhiko Arase, “QR Code Generator Library,” [Online]. Available: <https://www.npmjs.com/package/qrcode>.